

ARUNDHATI SENGUPTA

Computational Linguist — AI Training & Evaluation Data (Speech & Text)

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SUMMARY

Computational linguist with 10+ years running human data programs for language technology: gathering requirements, writing annotator instructions, defining quality criteria, training annotators and delivering quality-controlled speech and text data across 22 languages and scripts. Led end-to-end corpus programs at the Central Institute of Indian Languages (CIIL) for seven years, evaluated speech data for Google, and benchmarked four frontier LLMs on Bengali pronunciation. Ph.D. in Linguistics (CUNY).

CORE SKILLS

Program & Operations: Requirements gathering, annotator instructions and success criteria, annotator training, process improvement

Data Annotation & Quality: Guideline authoring, QC, inter-annotator agreement, adjudication, data quality audits, A/B testing, corpus curation

Large Language Models: Prompt engineering (zero-/few-shot), API-based LLM benchmarking (Gemini, Claude, GPT-4o), LLM error analysis

Speech Data (Multimodal): Speech corpus design and collection, acoustic and phonemic model evaluation, phonemic inventories, pronunciation lexicons (G2P), forced alignment (Praat/TextGrid)

Python & NLP: Python (API scripting, PyTorch, Hugging Face, scikit-learn, Pynini), SQL, R, UNIX, Git, finite-state (WFST) rules

Languages: English (native), Bengali (native), Hindi (professional)

EXPERIENCE

Ph.D. Dissertation Completion | Graduate Center, CUNY

Oct 2025 – Sep 2026

- Completed dissertation research on Bengali pronunciation modeling (approved Sep 2026), including the first multi-source Bengali G2P corpus and the first publicly available Bengali diacritization corpus.

Research Assistant — Language Data & Model Evaluation | Research Foundation of CUNY

Aug 2023 – Sep 2025

- Built and quality-controlled large multilingual annotated corpora with five native-speaker annotators across three languages; **wrote the annotation guidelines, trained annotators** and adjudicated disagreements so labels stayed consistent (BSF grant).
- Benchmarked four frontier LLMs** (Gemini 2.5 Pro, Claude Opus 4.5, Claude Sonnet 4, GPT-4o) on Bengali pronunciation. Even the best model had 2.5× the word error rate of a purpose-built neural model (39.5% vs. 15.8%), failing on sounds the spelling does not show.
- Built G2P and diacritization models in **Python/PyTorch** and integrated multilingual datasets with SQL (PSC-CUNY grant), reaching a **10% relative error reduction** over baseline.

Teaching Assistant, NLP-focused courses | Graduate Center, CUNY

Sep 2019 – Dec 2021

- Taught and led practicum sessions for Computational Linguistics Methods I and II, covering Python for NLP and text classification.

Freelance Linguistic QA Reviewer | Lionbridge Technologies, Inc.

Sep 2013 – Aug 2016

- Reviewed multilingual software output for linguistic quality, giving structured, actionable feedback to engineering and localization teams.

Speech Data Evaluator | Google Inc.

Nov 2012 – Aug 2013

- Evaluated acoustic and phonemic model performance on large-scale speech data; designed phonemic inventories and reported findings to engineering teams, improving models and training data.

Linguistic Data Scientist | Central Institute of Indian Languages (CIIL)

Oct 2005 – Oct 2012

- Designed and ran speech and text data collections** across 22 Indian languages and dialects, owning collection protocols, annotation guidelines and evaluation metrics through to **final delivery** of multilingual, multi-script language resources.
- Hired, trained and managed the LDC-IL team spanning data collection, annotation, UX and technical writing; directed prototype builds and benchmarked interfaces to choose the final design.
- Partnered with linguists, annotators and engineers as their liaison, turning quality findings into guideline and pipeline improvements; edited *Proceedings of Multilingual Speech Resources* (2010).

SELECTED PROJECTS

- G2P Error Analysis Tool (open source, github.com/arundhati-sengupta/g2pErrorAnalysis)** — WFST/rule-based QA tool that automates quality control and systematic error reporting on model output.

EDUCATION

Ph.D. in Linguistics, Graduate Center, CUNY — Dissertation: *A Study of Bengali Grapheme-to-Phoneme Conversion*

Aug 2019 – Sep 2026

M.A. in Linguistics, Graduate Center, CUNY — computational thesis: *Personality Trait Prediction from Speech*

2016 – 2018

M.A. in Linguistics, University of Calcutta, India | **B.A. in Linguistics**, University of Calcutta, India

2000 – 2005

SELECTED PUBLICATIONS

- Ashby, L. F. E., Gorman, K., ... **Sengupta, A.**, et al. (2021). Results of the Second SIGMORPHON Shared Task on Multilingual Grapheme-to-Phoneme Conversion. In *Proceedings of the 18th SIGMORPHON Workshop*, 115–125. ACL. My contribution: built the error analysis tool.
- Sengupta, A.** (2016). Voice Onset Time and Burst Duration of Bangla Labial Stop Consonants in CV Context. *Language in India*, 16(3), 37–45.
- Assistant Editor, *Longman-CIIL Basic English-English-Bengali Dictionary* (Pearson/Longman); Editor, *Proceedings of Multilingual Speech Resources*, CIIL, 2010.